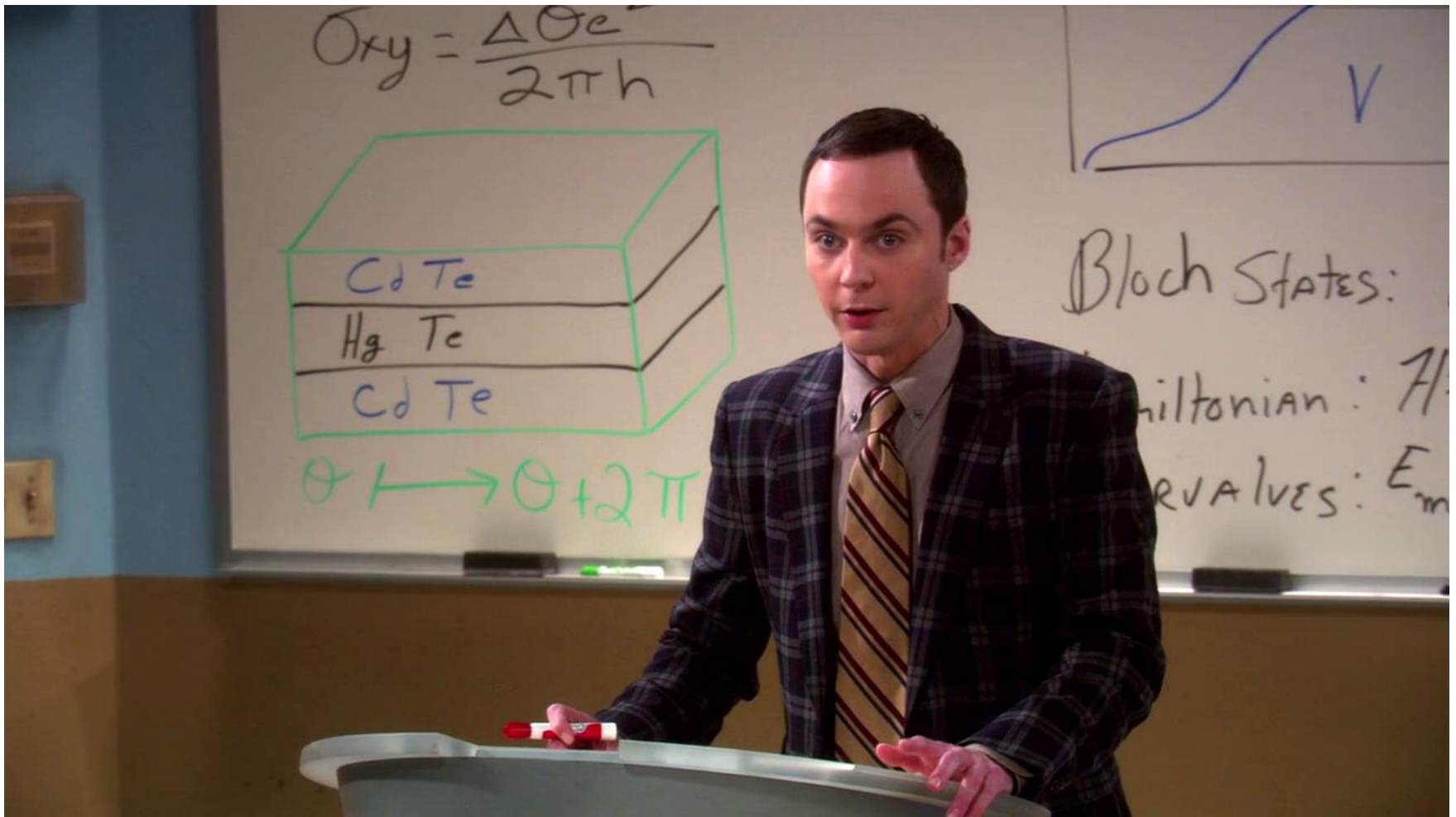
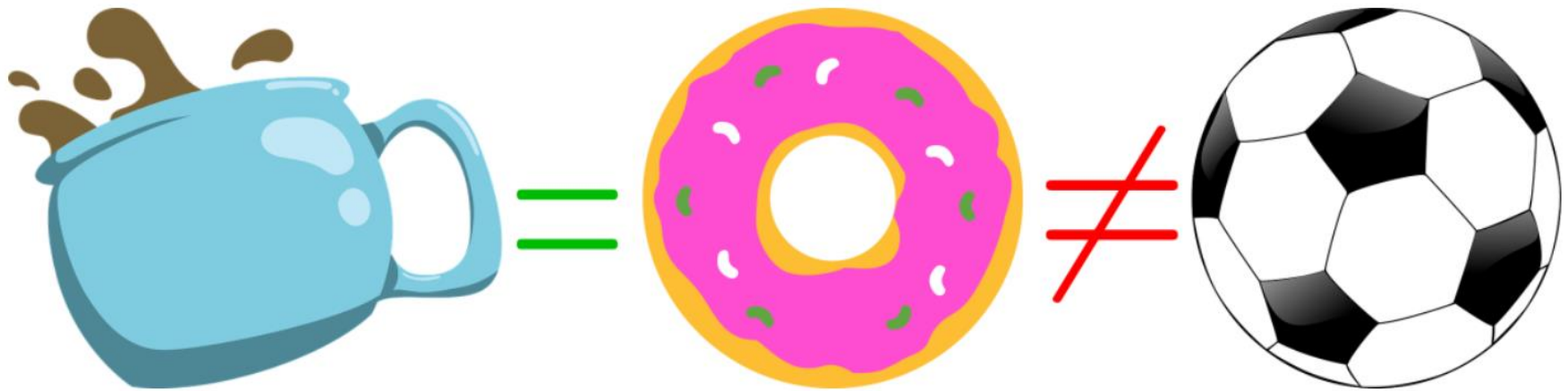


“ Who here is familiar with the concept of topological insulators? ”



Topology And Material Science

& some other interesting things



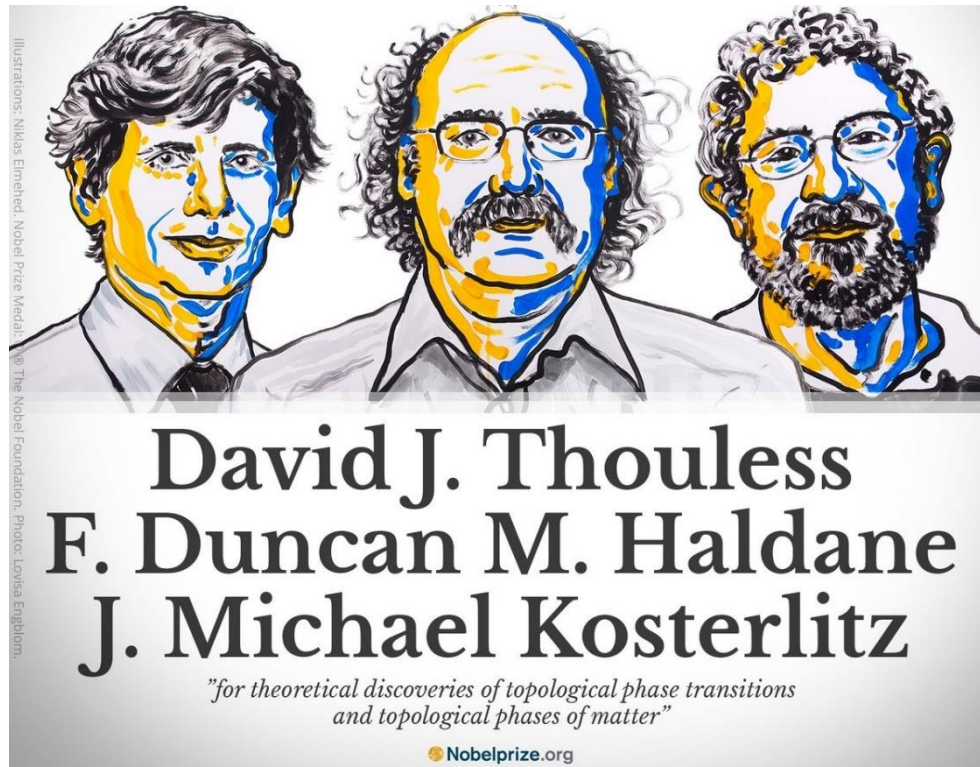
NIVEDITA
160462
nivdita@iitk.ac.in
For COM300A

Motivation

What if electricity could travel forever without being diminished? Computers running exponentially faster with unparalleled accuracy?



We might have an answer to what technology can help achieve this.

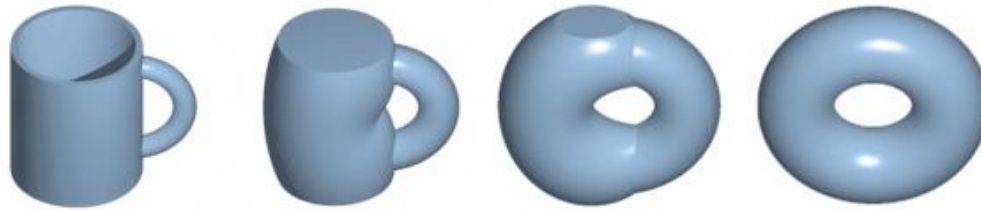


Topology

Topology is concerned with the fundamental properties of space and of objects that are preserved under continuous deformations, such as stretching, twisting, crumpling and bending, but not tearing or gluing.

Q: Who is a topologist?

A: Someone who cannot distinguish between a doughnut and a coffee cup.



Topological invariants are something that **characterize** spaces, we may smoothly deform one space into another.

All the surfaces which can be deformed into each other with such restrictions, we'll say, belong to the same *topological class*.

Example: A donut belong to the same class (1 hole), which is a class different from the class where a football (0 hole) would fall into, or say a pretzel (3 holes).

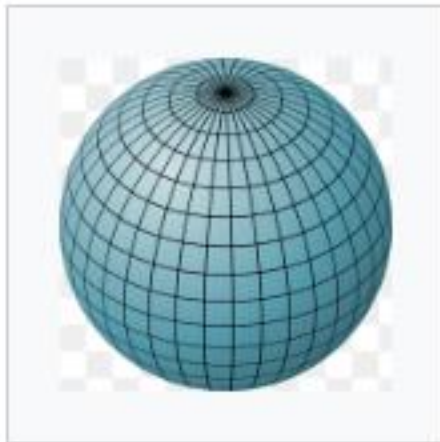
So, how do we mathematically define this notion of “hole”?
The answer is **Genus**.

Notion of Genus

The genus is a topologically invariant property of a surface (that can not be eliminated by any smooth deformation) defined as the largest number of **nonintersecting simple closed curves** that can be drawn on the surface without separating it.

OR

The maximum number of cuts along non-intersecting closed simple curves without rendering the resultant manifold (generalization of a surface, locally embeddable in $\mathbb{R}^n$) disconnected.



genus 0



genus 1



genus 2



genus 3

Gauss-Bonnet Theorem

The integral of Gaussian curvature k_g over the closed surface produces

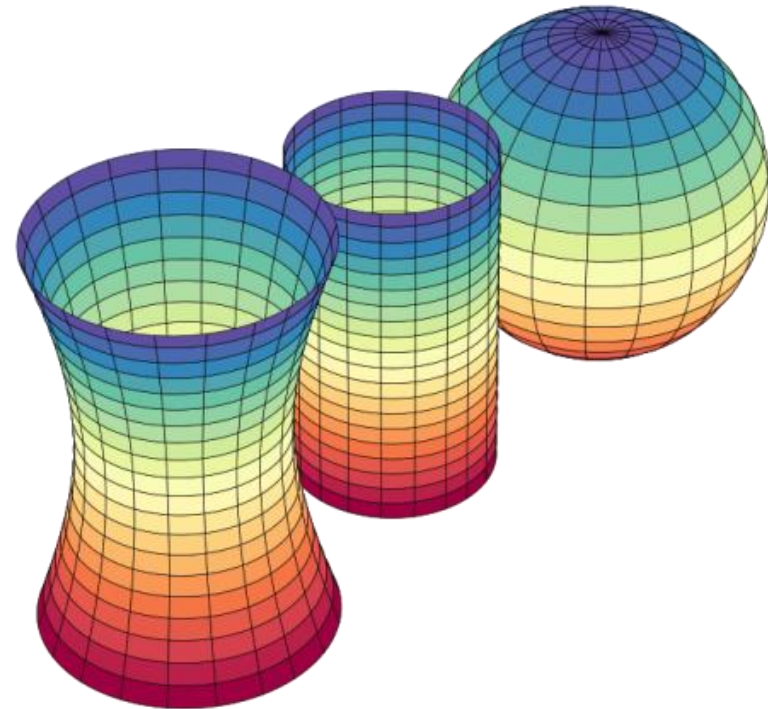
$$\oint_{surface} k_g \frac{dS}{2\pi} = 2 - 2g$$

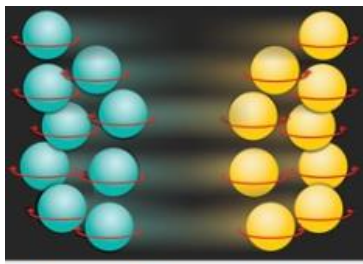
Where g is the genus, which characterizes the topology of the surface

The Gauss-Bonnet theorem or Gauss-Bonnet formula in differential geometry is an important statement about surfaces which connects their geometry (in the sense of curvature) to their topology.

Gaussian curvature of a surface at a point is the product of the principal curvatures, κ_1 and κ_2 , at the given point:

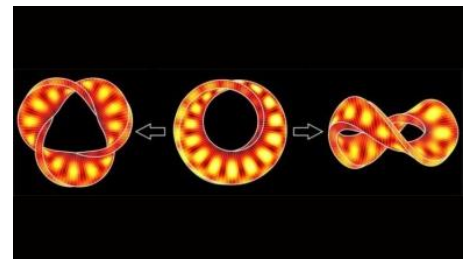
$$k_g = \kappa_1 \kappa_2$$





Novel Topological
Quantum Matter

Math to Matter



How **electrons** travel and occupy the solid is complicated, but they often have their own topological invariants.

If the topological invariant is **nontrivial** (non-zero), then we have examples like topological insulators, topological superconductors, quantum hall effect etc.

Materials can reach unusual regimes where their electrical properties present exotic behaviours, like **dissipation-less currents** and **quantized electrical resistance**.

The topological states are described by “magic” (topological) numbers.

The outstanding fact that quantum states of matter can be associated with topological numbers guarantees the robustness of their unique electrical properties against perturbations.

This suggests numerous promising technological applications, e.g. in spintronics and quantum computation, hence motivating the search for novel topological states of matter in laboratories.

Quantum Hall Effect

Closed surfaces/ compact manifolds $\leftrightarrow$ *Brillouin zone*, for the case of lattices.
Berry curvature that can be defined in the Brillouin zone.

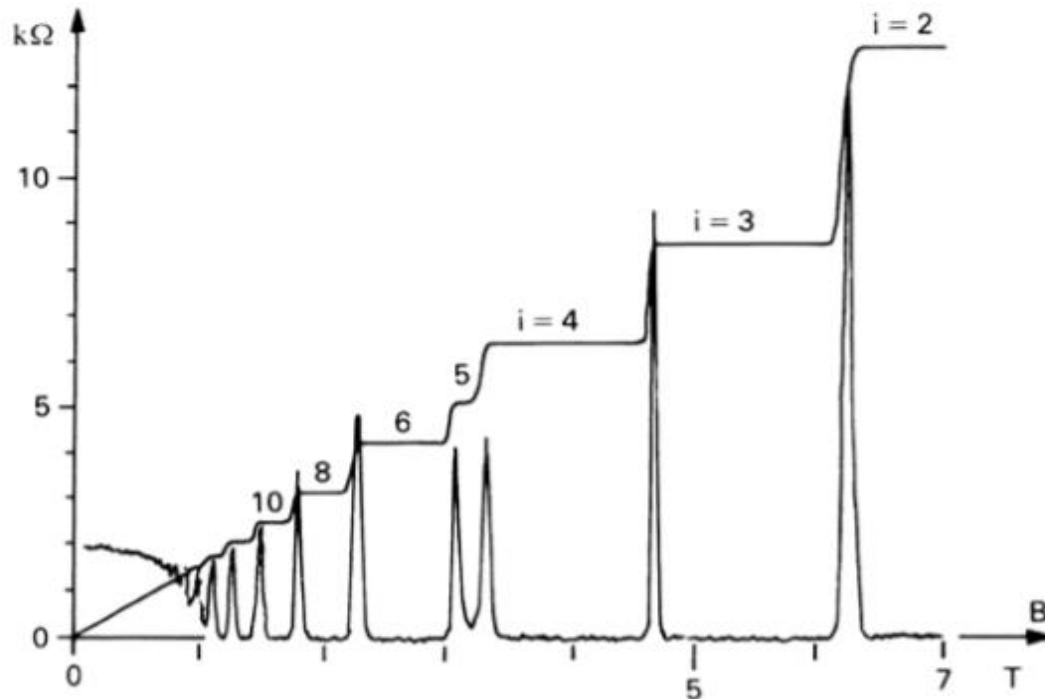
And it turns out that, the integral of the Berry curvature (in two dimensions) is something related to the Hall conductance σ_{xy} of a system.

$$\oint_{\text{Brillouin zone}} F_{xy} \frac{dk_x dk_y}{2\pi} = \sigma_{xy}$$

This is where the topology of the system comes into place. Because this integral needs to be an integer (times the unit of quantum conductance). This integer Hall effect is called the quantum Hall effect.

We find that the resistivity $\rho_{xy} = 1/\sigma_{xy}$, we get only values $\propto 1/\text{integers}$.
This was precisely what has been observed experimentally!

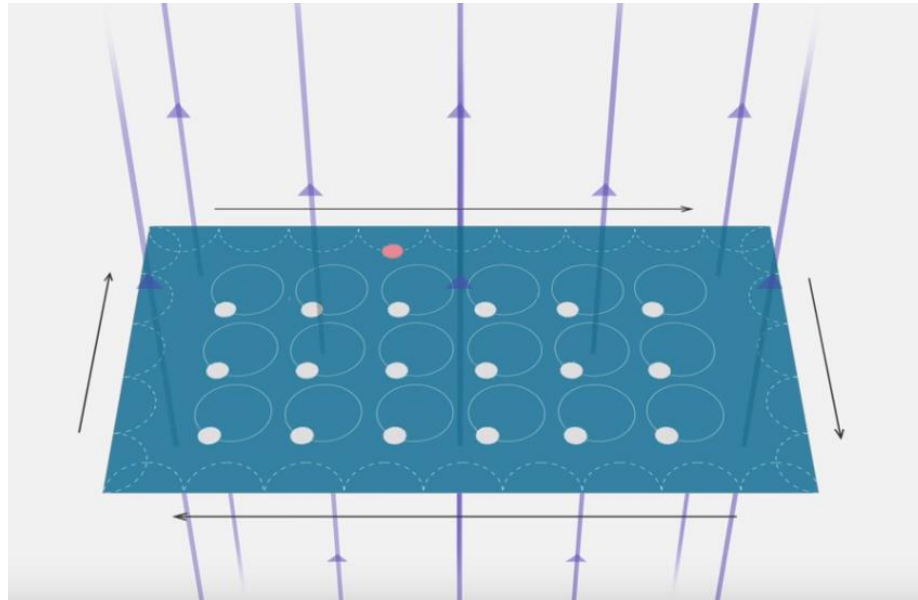
Quantum Hall Effect (Cont.,)



$$\rho_{xy} = \frac{2\pi\hbar}{e^2} \frac{1}{\nu} \quad \nu \in \mathbf{Z}$$

The value of ν is measured to be an integer to an extraordinary accuracy — $\sim 10^9$
Von Klitzing (Nobel Prize 1985)

Topological Insulators

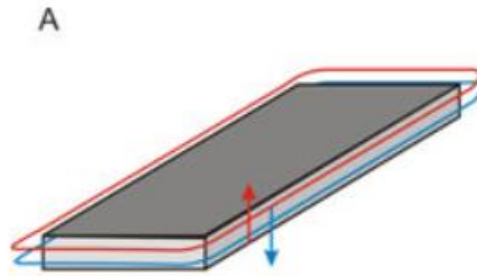


Imagine a thin layer of electrons, in a strong enough magnetic field.

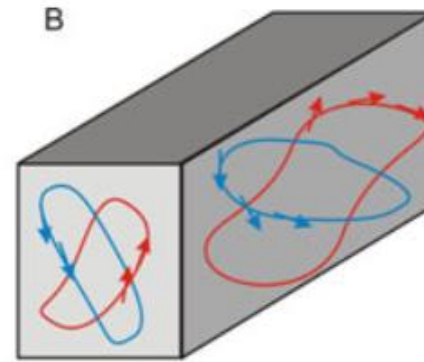
Each electron starts travelling in closed orbit and because electrons are stuck in these loops they don't conduct. These loops and their periodicity are topological properties of the system.

But, at the edge of the material, the orbits become open and connected. The electrons travel from one orbit to next and travel along the edge.

Here, we have a system conducting along edges and insulator in the bulk.

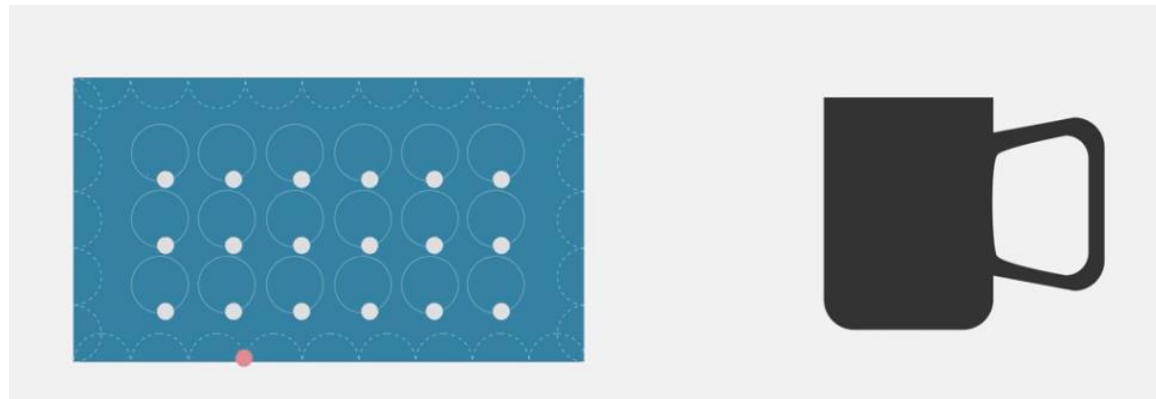


2D topological insulator



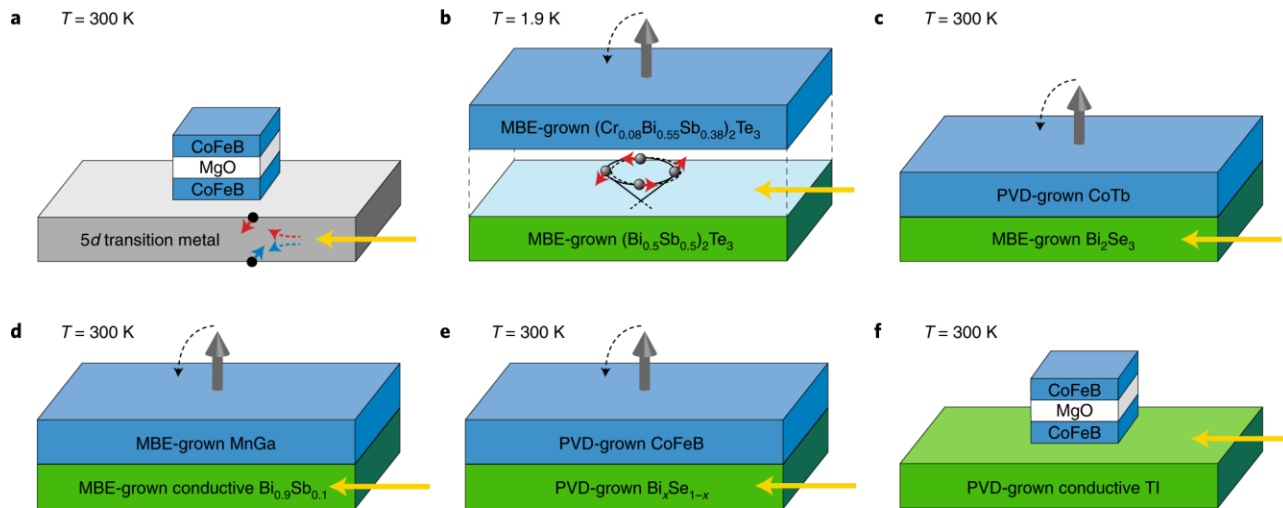
3D topological insulator

What is special about topological insulators is that their surface states are protected, the conductivity doesn't change by small imperfections or impurities (just like the hole in the mug doesn't change by stretching it).

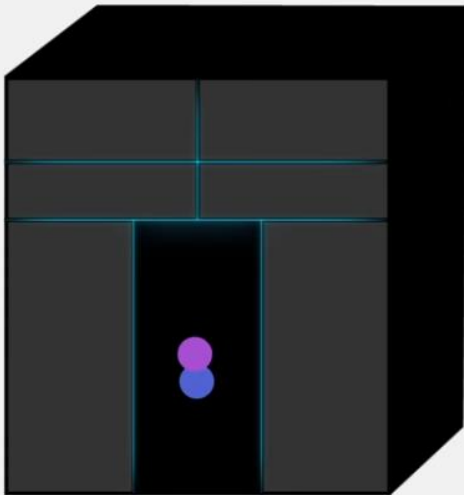


The edge states have “perfect” electron transport.

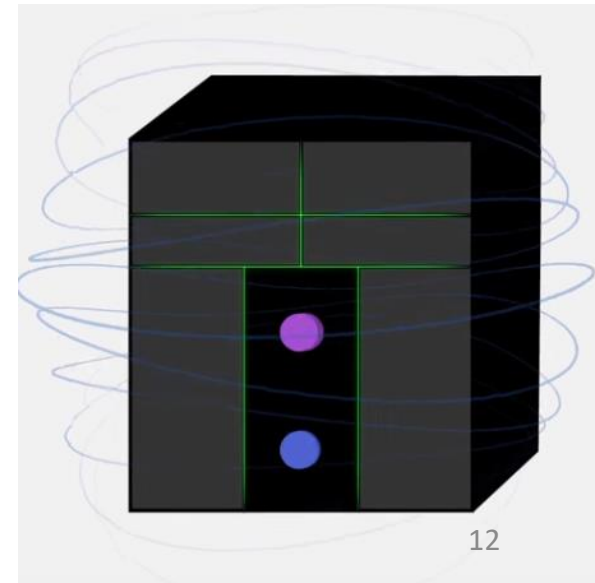
- No electrons travel backwards
- No energy is lost as heat
- No of conducting pathways can be controlled
- Future electronics can be built to exploit this perfectly efficient highway



HgTe, BiSb, BiSe are some other examples and newer materials are being explored to make bring topological materials into play for a revolutionary technology shift in electronics and computing.



Qubits formed by sub atomic particles which are **topologically protected** and are not changed by small or local disturbances, hence are more stable, leading to more accurate computations and a better Quantum computer.

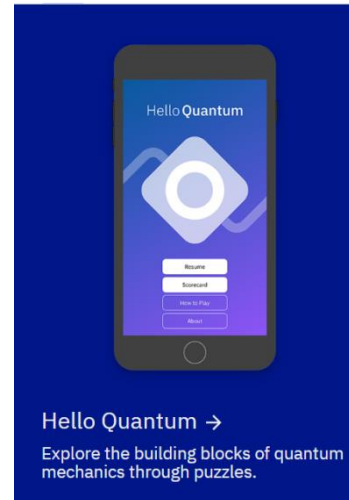


Interested?

Check out IBM's website on Quantum Computing and explore the world of Quantum Mechanics .

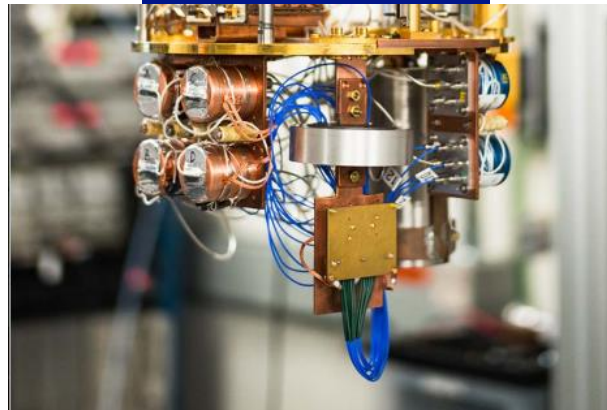
After all it all breaks down to logic (gates), circuits and materials to implement those.

IBM Q



Hello Quantum →

Explore the building blocks of quantum mechanics through puzzles.



That's all from my side.

Questions? Discussion?

